North Bengal International University

Lab Report: 04

Exp Name : Implementation of Signed Binary Multiplication Using Bit-wise Operator

Subject : Computer Architecture & Organization Sessional

Subject Code : CSE 0612-3142

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Experiment No: 04

Experiment Name: Implementation of Signed Binary Multiplication Using Bit-wise Operator in C Language

Theory: Signed binary multiplication involves multiplying two numbers (positive or negative) in binary format. Computers use **two's complement** to represent negative numbers. Instead of using the built-in `*` operator, we can simulate multiplication by checking each bit of the multiplier and adding the shifted multiplicand when necessary. Bitwise operators like `<<`, `>>`, and `&` help in performing this multiplication, which closely mimics how hardware arithmetic units work internally. This is an essential concept for understanding how basic operations are performed in computer architecture.

Code:

```c
#include <stdio.h>

int main() {
    int a, b, result = 0;
    int negative = 0;

    printf("Enter first number: ");
    scanf("%d", &a);

    printf("Enter second number: ");
    scanf("%d", &b);

    if (a < 0) {
        a = ~a + 1;
        negative = !negative;
    }

    if (b < 0) {
        b = ~b + 1;
        negative = !negative;
    }

    while (b != 0) {
        if (b & 1) {
            result += a;
        }
        a <<= 1;
        b >>= 1;
    }

    if (negative) {
        result = ~result + 1;
    }

    printf("Product is: %d\n", result);

    return 0;  }
```

Output:

```
Enter first number: -7
Enter second number: 4
Product is: -28
```

Conclusion: This lab showed how signed multiplication can be done using bitwise operations, helping us understand low-level binary arithmetic in computer architecture.